

Grammatical Metaphor Increases AI-Animacy Perception in Online Sentence Processing

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BACKGROUND

- Subject position of transitive clauses carries proto-agentive entailments, such as causativity and animacy.^{1,2}
- English allows this agentive subject position to be occupied by non-animates in a phenomenon known as grammatical metaphor.³
- Grammatical metaphor (GM)** increases the responsibility assigned to non-animates in rating tasks.^{4,5,6}
- It is not known whether **GM** relies on anthropomorphic beliefs.

RESEARCH QUESTIONS

- How does **GM** affect online sentence processing
- Does **GM** increase perception of AI animacy?
- How does the effect of **GM** interact with individual differences in anthropomorphism?

HYPOTHESIS

Grammatical metaphor will increase acceptance of AIs as subjects of animate verbs for participants with anthropomorphic beliefs

EXPERIMENT DESIGN

Independent Variables

- Framing Condition: **GM** / **no GM**
- Explanation Verb: **Animate** / **Inanimate**
- Technology Type: LLM / other AI / non-AI
- Outcome: Positive / Negative

Participant Variable:

- technology-Individual Differences in Anthropomorphism Questionnaire (**t-IDAQ**) (Modified version of the IDAQ,⁷ focused on AIs, LLMS, and Computer Programs)

STIMULI

36 Critical Items, 24 fillers (Ex. 2 Only)

GM

A text processing A.I. called **Lo-Go™** massively improved **Quantum Mutual Funds** popularity after they launched it. **Lo-Go™** generated summaries of thousands of earnings calls... in the first three months, **Lo-Go™** lost billions for investors.

no GM

By launching a text processing A.I. called Lo-Go™, Quantum Mutual Funds massively improved their popularity. By using Lo-Go™, Quantum Mutual Funds generated summaries of thousands of earnings calls... with Lo-Go™, investors lost billions in the first three months.

Animate Explanation

It all happened because Lo-Go™ had [judged] the earnings calls incorrectly.

Inanimate Explanation

It all happened because Lo-Go™ had [decoded] the earnings calls incorrectly.

[] marks the critical region for eye-tracking measures

METHODS & PARTICIPANTS

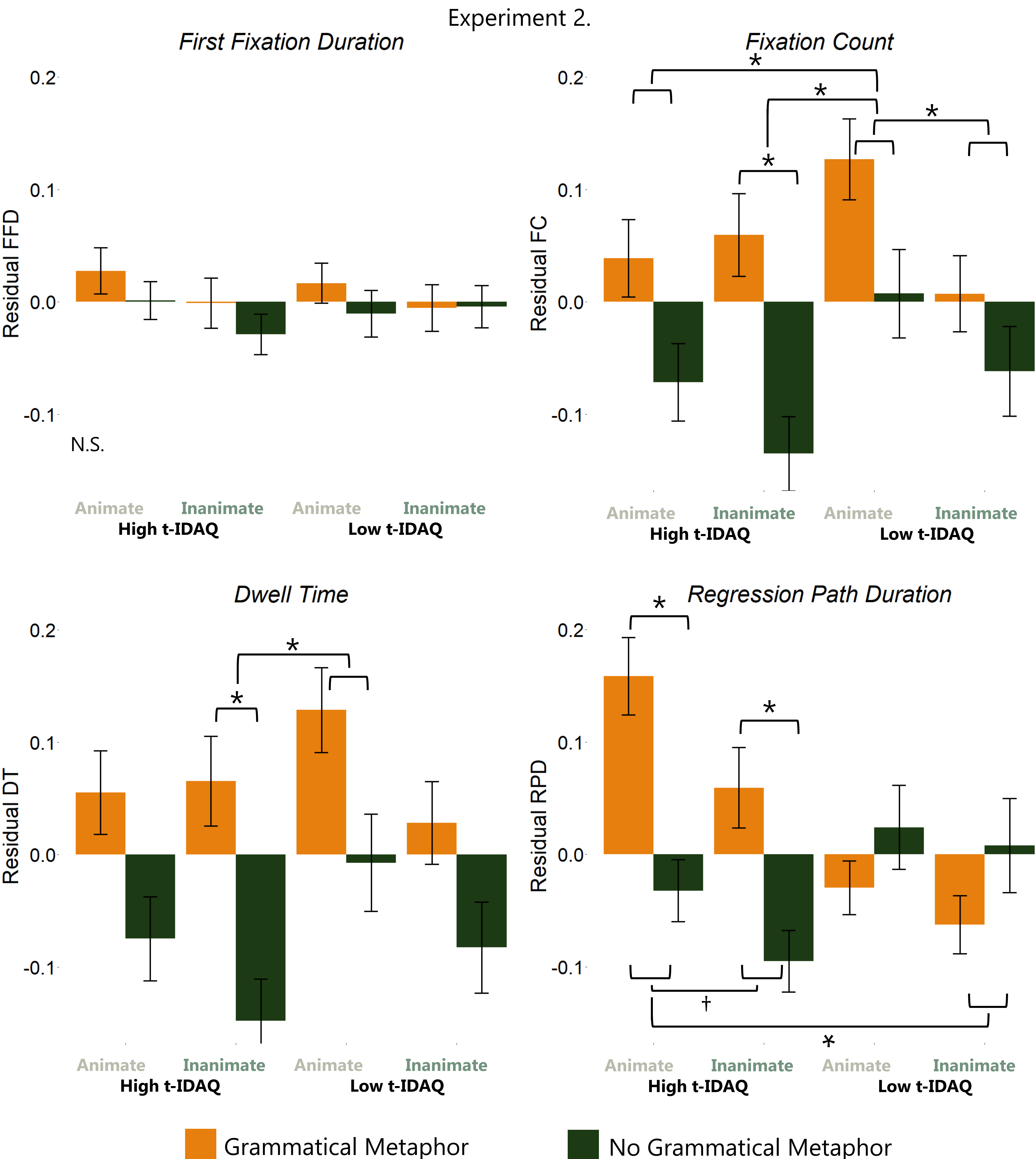
Experiment 1 (Ratings)

- 73 Participants recruited through Prolific
- Randomly assigned to **GM** / **no GM** framing
- Saw 18 items in each explanation condition
- Rated their agreement with the explanation

Experiment 2 (Eye-Tracking)

- 62 Participants recruited from UofSC
- Randomly assigned to **GM** / **no GM** framing
- Saw 18 items in each explanation condition + 24 filler items
- Eyelink 1000 used to determine reading times on critical explanation verbs

EYE-TRACKING RESULTS



ANALYSIS

Experiment 1

- Ratings modeled using ordinal regression.^{7,8}

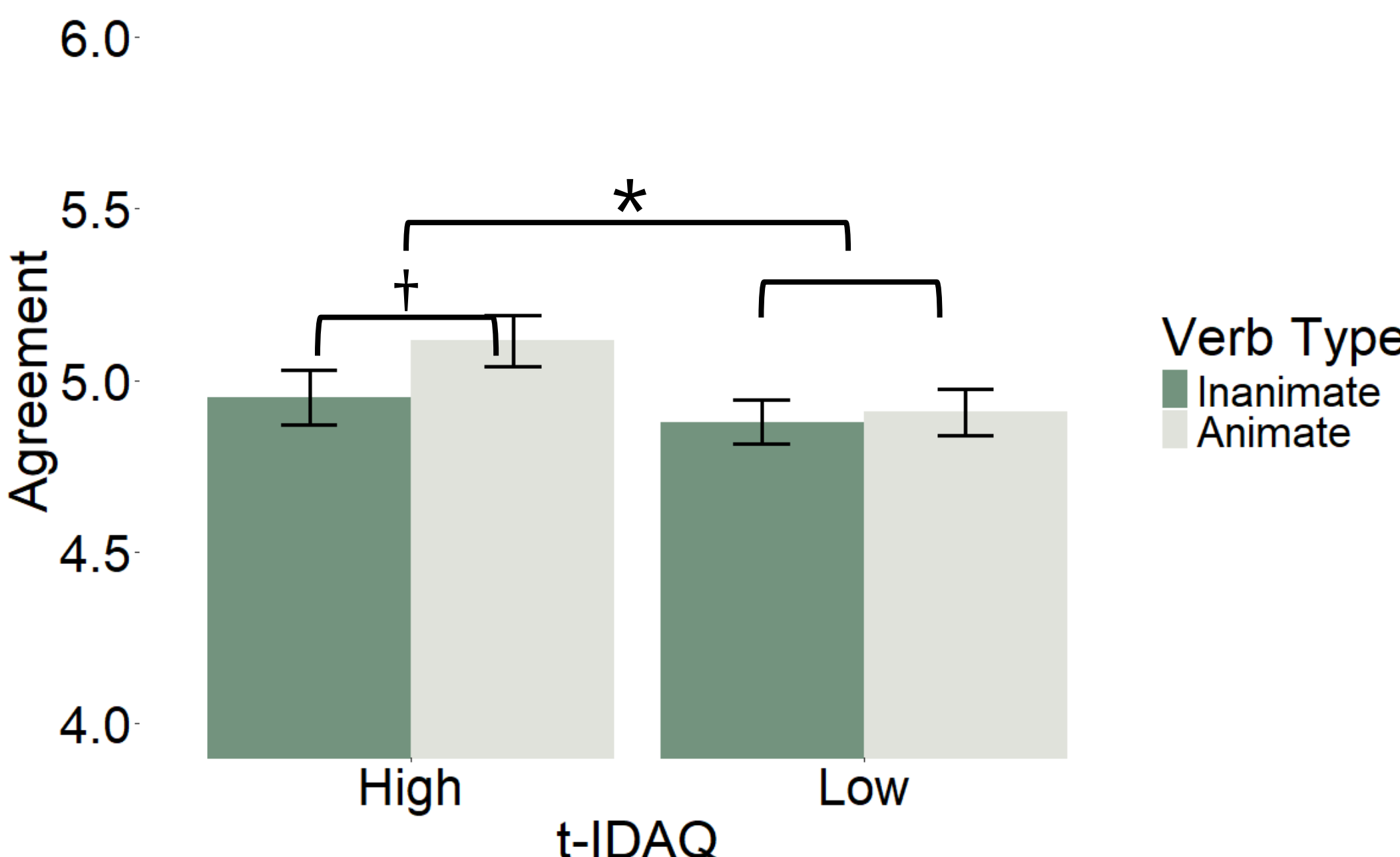
Experiment 2

- Data cleaned manually to correct vertical tracking errors
- First Fixation Duration, Fixation Count, Dwell Time, and Regression Path Duration on critical verbs were calculated as DVs
- All DVs were residualized using verb frequency, concreteness, and length
- Residuals were modeled with Participants, Items, and Trial Order as random factors

BEHAVIORAL RESULTS

Experiment 1.

Agreement Ratings



DISCUSSION

- We found an interaction between **Grammatical Metaphor** and **Explanation Verb Animacy**
- On middle/late processing measures (FC, DT, RPD)
- But **not** on the outcome measure (Ratings) or the early processing measure (FFD)
- The interaction depended on participants' anthropomorphic beliefs about technology (t-IDAQ)
- The interaction was not entirely in the direction predicted
- High t-IDAQ participants showed a compatibility effect only in the **no GM** condition, which was reduced by **GM**
- Low t-IDAQ participants showed a cumulative inhibitory effects from **GM** and **Animacy**
- The latest process measure (RPD) patterned differently from FC and DT
- High t-IDAQ participants showed a cumulative inhibitory effects from **GM** and **Animacy**

References

- ¹Dowty, D. (1991). Thematic proto-roles and argument selection. *Language*, 67(3), 547–619.
- ²Fausey, C. M., & Boroditsky, L. (2010). Subtle linguistic cues influence perceived blame and financial liability. *Psychonomic Bulletin & Review*, 17(5), 644-650. <https://doi.org/10.3758/PBR.17.5.644>
- ³ Devrim, D. Y. (2015). Grammatical metaphor: What do we mean? What exactly are we researching? *Functional Linguistics*, 2(1). <https://doi.org/10.1186/s40554-015-0016-7>
- ⁴Dragojevic, M., Bell, R. A., & McGlone, M. S. (2014). Giving radon gas life through language: Effects of linguistic agency assignment in health messages about inanimate threats. *Journal of Language and Social Psychology*, 33(1), 89-98.
- ⁵McGlynn III, J., & McGlone, M. S. (2019). Desire or disease? Framing obesity to influence attributions of responsibility and policy support. *Health Communication*, 34(7), 689–701. <https://doi.org/10.1080/10410236.2018.1431025>
- ⁶Petersen, D. & Almor, A. (In Press). Linguistic framing affects moral responsibility assignments towards AIs and their creators. *Frontiers in Psychology*.
- ⁷Waytz, A., Cacioppo, J., & Epley, N. (2010). Who sees human? The stability and importance of individual differences in anthropomorphism. *Perspectives on psychological science*, 5(3), 219-232.